

RECRUITING THROUGH THE STAGES: A META-ANALYTIC TEST OF PREDICTORS OF APPLICANT ATTRACTION AT DIFFERENT STAGES OF THE RECRUITING PROCESS

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We used meta-analysis and semipartial correlations to examine the relative strength and incremental variance accounted for by 7 categories of recruiting predictors across multiple recruitment stages on applicant attraction. Based on 232 studies (250 samples, 3,518 coefficients, $n = 108,632$), we found that characteristics of the job, organization, and recruitment process, recruiter behaviors, perceived fit, and hiring expectancies (but not perceived alternatives) accounted for unique variance in applicant attraction at multiple stages. Perceived fit was the strongest relative and unique variance predictor of applicant attraction albeit a nonsignificant predictor of job choice. Although not among the largest zero-order predictors, recruiter behaviors accounted for substantial incremental variance at the first 2 stages. Organizational characteristics are more heavily weighed by applicants when maintaining applicant status as compared to the stage of application, and recruitment process characteristics are weighed progressively more as the recruitment stages advance. Job characteristics accounted for the greatest unique variance in job choice decisions. Job characteristics are more predictive in field studies, whereas recruiter behaviors, recruitment process characteristics, hiring expectancies, and perceived alternatives produced larger effect sizes in the laboratory. Results are discussed in terms of their theoretical and practical implications with future research suggestions.

Demographic and economic changes over the past 30 years have led to a competition for qualified candidates referred to as a “war for talent” (Michaels, Handfield-Jones, & Axelrod, 2001). In light of recent economic setbacks, recruiting and retaining the right talent to meet organizational

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objectives in lean times may be of even greater import. The recruitment process in the next decade and beyond will be about providing prospective talent with the information they desire to make decisions about their job pursuit and job choices (Murphy & Tam, 2004), and striking the balance between an adequate supply of qualified talent and avoiding the costs associated with generating too many or unqualified applicants (Dineen, Ling, Ash, & DelVecchio, 2007).

Considerable research has been conducted to examine the factors that predict applicant attraction to work opportunities including job and organizational characteristics (e.g., Schwab, Rynes, & Aldag, 1987), recruiting practices (e.g., Dineen et al., 2007; Taylor & Bergmann, 1987) applicant perceptions of fair treatment (Chapman, Uggerslev, & Webster, 2003), fit (Dineen, Ash, & Noe, 2002), alternative employment opportunities (Bauer, Maertz, Dolen, & Campion, 1998), and hiring expectancies (Rynes & Lawler, 1983). This body of work, however, has been widely criticized for its lack of practical relevance (e.g., Breugh, 2008; Ployhart, 2006). Saks (2005) stated that "it is fair to say that (a) there are few practical implications for recruiters and organizations, (b) the practical implications that can be gleaned from recruitment research have been known for more than a decade, and (c) the main practical implications are at best obvious and at worst trivial" (p. 69). This paper seeks to answer the call for significant recruitment implications by using multiple regression and cell group moderator analyses on meta-analytically derived coefficients to address three longstanding questions. First, are certain categories of recruiting predictors more effective at different points during the recruitment process? Second, what are the strongest or most meaningful predictors of applicant attraction and job choice? Third, to what extent do predictor categories account for incremental variance in applicant attraction and job choice above and beyond other predictors? From a practical standpoint, one may reword these questions to be: (a) should organizations change the way they allocate their recruiting dollars at different stages of the recruitment process? (b) where should an organization spend its first recruitment dollar?, and (c) where should an organization spend its subsequent (incremental) recruitment dollars? In addition, we seek to offer a theoretical rationale as to why recruitment information and interventions may vary in effectiveness within and across the recruiting stages.

Recruitment Predictors of Applicant Attraction

Breugh (1992) defined recruiting as "those organizational activities that (1) influence the number and/or types of applicants who apply for a position and/or (2) affect whether a job offer is accepted" (p. 4). Apart

from the specific activities an organization conducts (such as online applications, job fairs, and recruitment interviews), applicant attraction will be affected by job and organizational attributes (Rynes, 1991) and applicants' perceptions of their treatment during the recruitment process, alternative job options, as well as fit with the organization (Gilliland, 1993; Saks & Uggerslev, 2010).

We address seven categories of applicant attraction predictors that have commonly been explored in recruiting research (Barber, 1998). The first category—job characteristics (e.g., Cable & Judge, 1994)—refers to the favorability of the attributes associated with the position. The most common job characteristics examined are the work itself and compensation. The second category—organizational characteristics (e.g., Gatewood, Gowan, & Lautenschlager, 1993)—includes the image of the company, work environment, familiarity to the applicant, and its location and size. The third category—recruiter behaviors (e.g., Harris & Fink, 1987)—includes behaviors that recruiters may engage in during the recruitment process. Commonly examined behaviors include competence, personableness, trustworthiness, informativeness, and time spent recruiting (Barber, 1998).¹ The fourth category—recruitment process characteristics—includes the attractiveness and usability of various recruiting activities (e.g., Cober, Brown, Levy, Cober, & Keeping, 2003) and the extent to which recruiting messages are perceived as complete, realistic, timely, and credible (Breaugh, Macan, & Grambow, 2008; Cable & Yu, 2006). Recruitment process characteristics also include reactions to recruiting and selection procedures—typically focused on procedural justice (Gilliland, 1993; Ryan & Ployhart, 2000)—and applicant perceptions of the job relatedness, treatment, timeliness, consistency, and the opportunity to perform on selection tests (e.g., Chapman et al., 2003). The fifth category—perceived fit—includes applicant perceptions of how well their goals, values, and ideals suit the job (person-job fit) and organization (person-organization fit; e.g., Ryan, Horvath, & Kriska, 2005). The sixth category—perceived alternatives—refers to applicants' perceptions of the availability of alternative employment opportunities (Chapman & Webster, 2006). And the seventh category—hiring expectancies—refers to the applicants' perceptions of how likely they are to receive a job offer from the company (Saks & McCarthy, 2006).

In a previous recruiting meta-analysis, Chapman, Uggerslev, Carroll, Piasentin, and Jones (2005) examined the relationships between recruit-

¹Notably, we have not included recruiter demographics within the recruiter category. This decision was based on meta-analytic evidence provided by Chapman, Uggerslev, Carroll, Piasentin, and Jones, (2005) indicating that recruiter demographics did not significantly predict application attraction.

ing predictor categories and four indices of applicant attraction: job pursuit intentions, acceptance intentions, job/organizational attraction, and job-choice decisions. Chapman et al. (2005) examined a structural model among the outcome variables and found support for an intentions-mediated model akin to the theories of planned behavior (Ajzen & Fishbein, 1977) and reasoned action (Ajzen, 1991). Whereas the Chapman et al. (2005) focus was on zero-order relationships between recruiting predictors and outcomes and the relationship among recruiting *outcomes*, this meta-analysis provides a complementary examination of the interrelationships among the recruitment *predictors*, which allows for a comparison of predictor strength and incremental variance. In addition, this paper explores (a) recruitment as a process occurring over time by examining whether and the extent to which these relationships change as a function of recruitment stage, (b) novel zero-order relationships between recruitment predictors and applicant attraction, (c) effect sizes produced by field versus laboratory studies, (d) whether single source bias predicts relationship strength, and (e) manuscript publication status as potential moderator of relationship effect sizes.

Stages of the Recruitment Process

Barber (1998) divided the recruitment process into three stages—generating applicants, maintaining applicant status, and influencing job choices. Although Barber's taxonomy is widely used and positively accepted (Collins & Han, 2004), relatively little research has been conducted on the different stages of the recruiting process (Dineen & Soltis, 2011; Turban, Campion, & Eyring, 1995). That is, there is little discussion in the recruitment literature on how the pattern of predictor importance may change throughout the recruitment process.

Practically speaking, it is during the first two stages where recruiters have focused their efforts—both in providing job and organizational information to applicants and engaging in various recruitment activities with applicants and prospects (Dineen & Soltis, 2011; Van Hove & Lievens, 2009). Few recruitment efforts are targeted at influencing job choices given that (a) the applicant presumably has a substantial amount of information about the job and organization, has engaged with recruiters, and has been in the process long enough to inform their job choice decisions; and (b) candidates who are not persuaded to apply and stay in the applicant pool vanish from this third recruitment stage (Allen, Mahto, & Otondo, 2007). Thus, the first two stages are the dominant focus of recruitment drives (Breaugh et al., 2008). Ignoring the issues of timing and measurement during these stages may obscure important differences in the magnitudes of observed relationships (Gilliland,

1993). Although several recruiting studies have measured relationships at multiple points during the recruitment process (e.g., Bauer et al., 1998; Shantz, 2003), it is unknown whether some categories of recruiting information are more influential at different points during the recruitment process.

Focusing on the first two stages has implications for the types of outcome variables measured in recruitment research. Typically it is attitudes (such as organizational attraction) and intentions (such as job pursuit or acceptance intentions) that are measured during the first two stages, with behavioral measures of job choice decisions coming into play at the third stage. Consequently, there is substantially less behavioral data as compared to attitudinal or intentions-based data with respect to the recruitment process (Harold, Uggerslev, & Kraichy, *in press*).

One theory that may be particularly germane to understanding which recruiting information predicts applicant attraction and job choice behavior across the different stages of the recruitment process is the elaboration likelihood model of persuasion (Petty & Cacioppo, 1986a). The elaboration likelihood model has received considerable support within social psychology and marketing fields (Rucker & Petty, 2006; Sparks & Areni, 2002), and recently has been applied to recruiting (Buda & Charnov, 2003; Cable & Turban, 2001; Carless & Imber, 2007; Dineen et al., 2007; Jones, Schultz, & Chapman, 2006; Larsen & Phillips, 2002; Roberson, Collins, & Oreg, 2005; Uggerslev, Willness, & Fassina, 2009).

The elaboration likelihood model suggests that individuals process the messages they receive from various sources according to their ability and motivation to scrutinize the information (Petty & Cacioppo, 1986a; Petty & Wegener, 1998). A person's ability is affected by the opportunity to assess the quality of the information (Petty & Cacioppo, 1986b), as well as their cognitive ability (Petty & Wegener, 1998) and need for cognition (Buda & Charnov, 2003). Their motivation, on the other hand, will be affected by their desire to carefully scrutinize the content of messages, including seeing the task as important (Chaiken & Maheswaran, 1994) and personally relevant (Cacioppo, Petty, & Morris, 1983). When both ability and motivation are high—and therefore elaboration likelihood (EL) is high—persuasion occurs through a careful scrutiny of the quality of the messages received. Rather than scrutinize the caliber of the message arguments, applicants may rely on low EL mechanisms when ability, motivation, or both is lower (Petty & Briñol, 2008). Low EL mechanisms include simple cues such as credibility of the information source (Buda & Charnov, 2003), relying on other people's opinions (Chaiken & Maheswaran, 1994), or using sheer length of the message to assess the quality of the information. They may also include using heuristics or rules of thumb such as "good workplaces make attractive

advertisements” to judge information quality (Chaiken & Maheswaran, 1994; Jones et al., 2006).

Researchers in the persuasion field have found that the attitudes that people form under higher EL are likely to be persistent over time, resistant to change, and more predictive of behavior than attitudes formed under lower EL (Petty, Haugtvedt, & Smith, 1995). In addition, heuristic cues may not provide for any discernible persuasion when people have the ability and motivation to evaluate message quality (Chaiken & Maheswaran, 1994).

Both ability and motivation to scrutinize recruiting information for its evidentiary merit may be low before a person enters the applicant pool. Similarly, a potential applicant’s ability may be low because they have not had any exposure to acquire adequate information about the job and organization to scrutinize (Uggerslev et al., 2009). Therefore, applicants may be reliant upon simple cues from the recruiter (such as friendliness) or recruitment activities (such as a user-friendly website) when determining their attraction to an opportunity. In terms of motivation, applicants at early stages may be skimming or loosely considering a number of recruitment opportunities (Jones et al., 2006) and their commitment and emotional attachment to any one opportunity may be low (Ryan & Ployhart, 2000). As such, at initial recruitment stages, both applicant ability (due to a lack of information) and motivation (due to the number of opportunities and lack of commitment to any one opportunity) may be low leading to lower EL and therefore reliance upon heuristics and simple associations between recruiters and applicant attraction.

Important, it is still possible that a threshold level of some job and organizational characteristics must be present in order for applicants to pursue an opportunity and continue in the stages of the recruitment process. For instance, there may be a minimum “reservation wage” (Lippman & McCall, 1976) or location (Barber & Roehling, 1993) for applicants to enter the applicant pool. Furthermore, a minimum level of attention must be garnered before job seekers can be attracted to an opportunity (Roberson et al., 2005) and become part of the recruitment process. Other job and organizational characteristics such as work/life balance, organizational image, familiarity, and the work itself, may become more important as the stages of the recruitment process proceed (Boswell, Roehling, LePine, & Moynihan, 2003). Of note, Jones et al. (2006) found some variability in job seekers’ levels of EL; thus, it is possible that some applicants may have higher levels of motivation and may have already obtained significant amounts of information about an opportunity through corporate websites or employee referrals even early on in the recruitment process.²

²The authors would like to thank an anonymous reviewer for this insight.

As applicants enter the applicant pool and decide whether to remain, presumably they will gain more information about the job and organization and therefore can rely more upon their scrutiny of the caliber of the information received when determining their attraction to an opportunity (Petty & Briñol, 2008). Furthermore, as applicants get closer to making job choice decisions, their motivation to engage in deeper, effortful scrutiny of the information that they receive may commensurately increase as well as their commitment and emotional attachment to the opportunity. They may also have a better sense of their employment prospects in other searches progressing concurrently (Chapman & Webster, 2006). Thus, applicant EL may go up as the recruitment stages advance because of increases in both ability and motivation to scrutinize the information received. The simple associations that applicants may have initially formed between recruiters and attraction may become almost nonexistent in terms of their persuasiveness at later stages when applicants systematically process message quality (Chaiken & Maheswaran, 1994). This suggests that job and organizational information as well as characteristics of the recruitment process (particularly those aspects related to procedural justice) may become more important, and recruiters may become less important as the recruitment process unfolds.

Image and signaling theories may lend some further support to understanding the relationship between recruitment information and applicant attraction. In image theory, Beach (1990; 1993; Beach & Mitchell, 1987) refers to the decisions made early on as a search for violations or reasons to screen out options from a final choice set. At this stage, applicants may have little information other than relying on early contact with organizational recruiters to use as signals of working conditions at the organization (Rynes, Bretz, & Gerhart, 1991). However, van Zee, Paluchowski, and Beach (1992) found that information that is used to make the early screen-out decision is not used again in making the final choice decision (see also Potter & Beach, 1994), and Rynes et al. (1991) found that signals may have little discernible impact once more information is known. Thus, information known early in the process (such as recruiter behaviors) may be used to make initial screening decisions. Applicant experiences during the recruitment process and specific information about job and organizational characteristics (such as salary information and the work environment) may inform later stage decisions.

To our knowledge, examining whether the stage of the recruiting process moderates the relationships between the predictor categories and attraction outcomes has not been empirically tested. Nevertheless, we contend that there is sufficient theoretical and empirical evidence to suggest that stage will impact relationship effect sizes. Ultimately, if we can ascertain the stage at which a recruiting predictor is most predictive of applicant attraction, then we can offer advice to practitioners regarding the

timing of their recruiting practices. In addition, we can help researchers to hone their examinations of various recruiting activities to specific stages of the process. Accordingly, we propose the following two hypotheses:

Hypothesis 1: Job characteristics, organization characteristics, recruitment process characteristics, and perceived alternatives will be more predictive of applicant attraction at later as opposed to earlier stages of the recruiting process.

Hypothesis 2: Recruiter behaviors will be more predictive of applicant attraction at earlier as opposed to later stages of the recruiting process.

Using the elaboration likelihood model to assess whether perceived fit and hiring expectancies may be more predictive at earlier or later recruitment stages is somewhat less straightforward. At early stages, perceived fit assessments may be made on the basis of heuristics or simple associations such as “the recruiter is similar to me, therefore I would like to work for this organization.” At later stages, perceptions of fit may be made on the basis of a careful scrutiny of all of the job and organizational information that has been received during the recruitment process and an assessment of how the quality of this information fits with the applicant’s needs (Rynes et al., 1991). Indeed, Roberson et al. (2005) found that perceptions of fit were higher when recruiting messages were detailed as opposed to general. Thus, perceptions of fit may be more accurate after a careful scrutiny of the job itself and the organization values, culture, and work environment (Cable & Judge, 1994). In terms of EL, both motivation (due to the desire to make an accurate fit assessment) and ability (due to the amount and specificity of information available) to make accurate fit assessments—and then to be persuaded by them—may go up as the recruitment stages progress. In short, although the mechanism of persuasion may change during the course of the recruiting process (from low to high EL), it is unclear how heuristics or careful scrutiny will affect the magnitude of relationship between fit with applicant attraction. Likewise for hiring expectancies, whereas early assessments may be made using simple heuristics and later assessments may be made on the basis of more evidence (e.g., passing selection testing; Chan, Schmitt, DeShon, Clause, & Delbridge, 1997), it is unclear how the changing mechanism will affect the magnitude of the hiring expectancy to attraction relationship. We therefore offer the following research question:

Research Question: Does the relationship between perceptions of fit or hiring expectancies with applicant attraction change as a function of the stage of the recruiting process?

Relative Strength and Incremental Variance of Recruitment Predictors

In their theoretical explanation of why various recruiting predictors might be more or less important to job choice decisions, Behling, Labovitz, and Gainer (1968) described three implicit theories. Under *objective factor* theory, applicants are thought to make job-choice decisions based on the attractiveness of job attributes. In *subjective factor* theory, applicants are assumed to seek jobs that fit their emotional and psychological needs. *Critical contact* theory suggests that applicants cannot make meaningful distinctions between jobs based on job attributes or fit with the opportunity, so they rely on critical contacts such as recruiters as the means of differentiation.

There are only a handful of studies that have attempted to compare predictor effectiveness and address the very practical question of where recruiters should invest their recruitment dollars. This paucity may be precipitated by issues with making fair relative comparisons (Cooper & Richardson, 1986) and what Wanous and Collela (1989) have described as a “contest” between job attributes and recruiting practices in terms of which is more important to applicants when making their job-choice decisions. Barber (1998) argued that although making fair relative comparisons between these predictors might be difficult, there is practical value in comparing the relative predictive strength and also determining the incremental variance of the recruiting predictors on applicant attraction. Meta-analytic procedures might be uniquely suited to making fair comparisons among these predictors because of the aggregate nature of the effect sizes and intercorrelations (Schmidt & Hunter, 2001; see also Connell & Goodman, 2002, and Orsingher, Valentini, & de Angelis, 2010, for meta-analytic relative strength comparisons). We contend that it is important to examine both relative strength and incremental variance in order to ascertain which of the recruiting predictors noted above can, in fact, predict applicant attraction uniquely and with practical significance.

On balance, the limited empirical evidence comparing recruiting predictor effectiveness suggests a direct relationship between job and organizational characteristics with applicant attraction but minimal or no support for the direct effects of recruiters or recruitment practices over and above job and organizational attributes. For instance, Rynes and Miller (1983) found that recruiter characteristics had an effect on job attraction when job attributes were held constant but not when job attributes were manipulated. Rynes and Miller also found that neither recruiter characteristics nor job attributes predicted acceptance intentions.

Although Rynes and Miller (1983) speculated that recruiter behavior might have a larger impact in a field setting with motivated job seekers, Powell's (1984) field study used a cross-sectional design with a

graduating college student sample and found that recruiter characteristics did not significantly predict acceptance intentions when controlling for job attributes. With similar result, Taylor and Bergmann (1987) used a correlational field design and found that recruitment activities were related to applicant reactions during the initial interview stage but not during later stages of the recruitment process, and not incrementally to organizational attributes. In addition, using structural equation modeling (SEM) on a field sample of on campus interviews, Turban, Forret, and Hendrickson (1998) found only an indirect relationship between recruiter behaviors and applicant attraction through influencing perceptions of job and organizational attributes.

For practitioners and recruitment researchers alike, it seems incomprehensible that recruiters may have little effect on applicant attraction through either their interactions with job seekers or their design and implementation of recruitment interventions. Indeed, the vast amount of money and effort that recruiters continue to pour into the training of recruiters (e.g., Sullivan, 2004), and the volume of research that continues to accumulate examining, for instance, the appeal of web sites (e.g., Dineen et al., 2007) and applicant reactions to the recruiting process (e.g., Saks & Uggerslev, 2010), suggests a disbelief that recruitment interventions have no effect.

Although the effect sizes have been small, several other studies have found recruiters to be a significant and/or direct predictor of applicant attraction above and beyond job and organizational attributes. Using a pre-post interview design in a field setting, Harris and Fink (1987) found that recruiter characteristics indirectly predict acceptance intentions through perceived job attributes and regard for the job and company but also directly predict acceptance intentions after controlling for job attributes. More recently, Carless and Imber (2007) used SEM and found that interviewer characteristics predicted applicant attraction directly and indirectly through job and organizational characteristics. In addition to recruiter characteristics, Powell (1991) found that recruitment practices predicted applicant reactions to an employment interview over and above job attributes using a pre-post interview design.

To understand the relative predictive strengths of the recruiting predictors on applicant attraction and job choice behaviors, the elaboration likelihood model and signaling theory may be able to shed some light (Petty & Cacioppo, 1981; Rynes et al., 1991). Although it is possible to scrutinize information gleaned from recruiter behaviors or characteristics of the recruitment process as arguments (Petty & Briñol, 2008), it is more likely that applicants will use this information to create simple associations or heuristics as demonstrated in advertising when viewers developing ideas about products based on the credibility of spokespersons

(Lutz, MacKenzie, & Belch, 1983). For example, when recruiters are seen as personable, competent, informative, and/or credible, applicants may tend to make fewer counterarguments to the claims made and therefore be more persuaded by the message (Heesacker, Petty, & Cacioppo, 1983). Similarly, an applicant may associate a functional recruiting website with organizational professionalism or associate a realistic job preview video depicting employees of multiple ethnicities with diversity as an important organizational value. Recruiter behaviors and characteristics of the recruitment process may act as signals from which the applicant can infer working conditions (Rynes et al., 1991).

Information about the job or organization, in contrast, is more likely to generate higher EL, as it conveys specific information about the employment opportunity (Petty & Wegener, 1991; Sparks & Areni, 2002). This is not to suggest that the simple presence of job and organizational information will generate high EL; applicants must also have the motivation to engage in effortful processing of information and the ability to do so (Buda & Charnov, 2003). As the personal relevance of a message increases, so does the time and effort spent on processing the message (Cacioppo et al., 1983). For instance, people who believe that an outcome will affect them personally are more affected by argument quality than the source or number of arguments (Petty & Cacioppo, 1984). Likewise, an applicant with higher EL is less likely to use the personableness of a recruiter as a simple cue about the working atmosphere within an organization as compared to examining the cogency of recruiter personableness to the opportunity. The applicant may decide that personableness is relevant if the recruiter is a future coworker but not relevant if the recruiter is from a human resources department.

Because recruitment contexts are likely to be high in personal relevance, applicants are likely to expend a great deal of effort processing job and organizational information. Moreover, because systematically processing the quality of messages leads to significantly stronger attitude persuasion than relying upon simple associations or heuristics (Chaiken & Maheswaran, 1994), job and organizational information may be stronger predictors of applicant attraction than characteristics of recruiters and the recruitment process.

To understand the relative predictive strength of recruitment predictors, Barber (1998) suggested the need to explore the interplay among predictor variables as all of the attributes and procedures in the recruitment process likely influence applicant's ultimate job-choice decisions. For example, fit perceptions are likely related to job and organizational characteristics because the favorability of these characteristics may inform candidate perceptions of how well they are suited for (or fit) with the job and with the organization (person-job and person-organization fit).

Therefore, perceived fit may be a larger predictor of applicant attraction and job choices than recruiter behaviors and characteristics of the recruitment process. Perceived alternatives and hiring expectancies have received relatively little attention (Chapman et al., 2005), and their relative strength and incremental variance compared to the other recruitment predictors is unknown. We therefore propose the following hypotheses:

Hypothesis 3: Job characteristics, organizational characteristics, and perceptions of fit will be stronger predictors of applicant attraction and job-choice behaviors than recruiter behaviors and characteristics of the recruitment process.

Hypothesis 4: Recruiter behaviors and characteristics of the recruitment process will account for unique variance in applicant attraction and job-choice behaviors above and beyond variance accounted for by job characteristics, organizational characteristics, and perceptions of fit.

Moderating Effects of Study Design, Rating Source, and Publication Status

Meta-analysis affords the opportunity to examine whether between-study differences may account for effect-size differences across primary studies (Schmidt & Hunter, 2001). In this study we explore three between-study differences: study design, rating source, and publication status. Within the recruitment field, one longstanding debate has surrounded the use of laboratory versus field designs (Barber, 1998). Issues with sample size, attrition, self-report and common method bias, and control of confounds have prompted researchers to use laboratory designs (Saks & Uggerslev, 2010). Saks and Uggerslev (2010) describe how laboratory designs also allow for controlled manipulations of recruiter behaviors and the recruitment process such as uninformative recruiters or delays in the recruitment process, which would be potentially unethical and contrary to human resource goals if conducted in the field. However, a thorough understanding of actual applicants and their reactions and behaviors in actual job search situations may optimally be captured through field designs (Breaugh, 2008). In this study, we test potential differences in effect sizes between laboratory and field designs to guide researchers in choosing future methodologies for their recruitment studies.

It is common for recruitment researchers to measure both perceptions of the predictor variable (e.g., attractiveness of job or organizational attributes, favorability of recruiter behaviors, reactions to recruiting activities) and criterion variable (e.g., job/organizational attraction, job pursuit intentions) using the same source (i.e., the applicant). As a form of

common method variance, single-source bias may inflate effect sizes rendering the true nature of the relationship (or nonrelationship) unknown (Podsakoff & Organ, 1986). In this study, we test whether relationship effect sizes are different when provided by single sources versus multiple or externally verified sources to guide researchers in choosing future data sources for their recruitment studies.

As a measure of primary study status, we also examine the methodological moderator of publication status. Specifically, we explore whether published studies produce different effect sizes than unpublished studies (e.g., Foldes, Duehr, & Ones, 2008; Gilboa, Shirom, Fried, & Cooper, 2008).

Method

Defining the Population of Studies

A comprehensive search for primary studies proceeded through three steps. First, a systematic search of PsycInfo and ProQuest databases from 1867 to June 2011 was conducted using the keywords personnel recruitment, recruitment, applicant, applicant attitudes/reactions, applicant attraction, job pursuit intentions, job pursuit behavior, application decisions, job organizational attraction, acceptance intentions, recruiter characteristics, recruiter behaviors, job characteristics, organizational characteristics, justice, fairness, perceived fit, person-job fit, and person-organization fit. Additional searches were conducted with a myriad of specific job, organizational, and recruitment process characteristic terms such as compensation, image, and job advertisement. Second, the reference sections of five seminal and recent reviews (Breugh, 2008; Chapman et al., 2005; Dineen & Soltis, 2011; Hausknecht, Day, & Thomas, 2004; Rynes & Cable, 2003) were examined to identify any studies that were not uncovered in the first step. Third, we examined recent conference programs (2003 to 2011) from the *Academy of Management* and *The Society for Industrial and Organizational Psychology* as sources of unpublished work.

Criteria for Study Inclusion

Studies had to meet four criteria to be included in the meta-analyses. First, the study context had to be one wherein participants were applicants engaged in one phase of the recruitment process.³ Second, the study had to contain data from the applicant's as opposed to the recruiter's

³When examining sample type, Chapman et al. (2005) found only found significant differences wherein real applicants weighed job characteristics and justice perceptions

perspective. Third, the study had to contain empirical data relating to one of the relationships included in the meta-analysis. That is, a coefficient for a relationship between (a) one of the seven a priori identified predictor categories and applicant attraction, or (b) two predictor categories to allow for tests of relative strength and incremental variance. Fourth, the sample size for each study needed to be greater than the number of coefficients contained in the study that would be coded for our analyses. Some studies that met the inclusion criteria were excluded because they did not meet the requirements for meta-analysis (e.g., they only provided partial or semipartial relationships; Hunter & Schmidt, 1990; Schmidt & Hunter, 2001). Applying the inclusion and exclusion criteria yielded 232 studies containing 250 samples with 3,518 coefficients for the meta-analyses. The total sample size from the studies was 108,632.

Meta-Analytic Approach

Coding of the Data

Two of the authors established the data coding scheme and completed independent codes of the first 95 studies. The kappa interrater agreement between the coders was 0.98. The primary coder completed the coding for the remaining studies using the established coding scheme and conversing with the second coder about any codes that were not clear cut. Previous recruiting research has been criticized for a lack of construct clarity, particularly with respect to the applicant attraction outcomes (Highhouse, Lievens, & Sinar, 2003). For the purposes of this meta-analysis, we have aggregated measures of job pursuit, acceptance intentions, and job/organizational attraction, and those with mixed content into one applicant attraction outcome for the following four reasons. First, there are very high correlations between the measures of applicant attraction: Chapman et al. (2005) found meta-analytically derived correlations ranging from 0.67 to 0.78 among job/organizational attraction, job pursuit intentions, and acceptance intentions, and Highhouse et al. (2003) found a correlation of 0.85 between attraction and intentions. Second, a number of recruiting studies have used measures that mix content between attraction and intentions measures (e.g., Carless & Wintle, 2007). Third, our focus is on

more than mock applicants specifically for the job–organization attraction outcome. Thus, consistent with Chapman et al. we included studies where applicants were actually involved in a recruitment process (i.e., authentic contexts) and studies where applicants were asked to pretend that they were involved in a recruitment process as typically occurs in laboratory settings (i.e., hypothetical contexts; Hausknecht et al., 2004). Studies of examinees who were asked to respond to selection tests without a recruiting context (e.g., Chan et al., 1997) were not included.

exploring the front-end relationships and not the relationships between outcomes. Finally, by aggregating across the attraction indices, we could obtain a substantially larger number of coefficients to explore different phases of the recruiting process as a moderating variable. We also coded job-choice behaviors as a separate outcome.

All predictors were categorized into one of the following categories: job characteristics, organizational characteristics, recruiter behaviors, recruitment process characteristics, fit perceptions, hiring expectancies, and perceived alternatives. To provide further specificity, we used content coding of the primary literature to indicate further categories and subcategories that can be meaningfully interpreted. Within job characteristics, we generated a further two categories (total compensation and job itself) and seven subcategories (compensation/pay/salary, benefits, advancement/promotions, development, autonomy, challenge, travel). The compensation/pay/salary subcategory included coefficients relating perceived favorability of pay (e.g., Van Hoye & Lievens, 2009), actual salary figures/compensation packages (e.g., Cable & Turban, 2003; Greening & Turban, 2000), and estimates of the pay to be provided by the job (e.g., Thorsteinson & Highhouse, 2003) with applicant attraction. Coefficients were classified into the benefits subcategory when they related perceived favorability (e.g., Kroustalis, 2009) and type of benefit package offered (e.g., Cable & Judge, 1994) to applicant attraction. For the job itself subcategories, autonomy referred to perceptions of freedom/autonomy in the prospective job (e.g., Slaughter & Greguras, 2009), advancement/promotions and development subcategories included the respective favorability of these opportunities (e.g., Lievens, Van Hoye, & Anseel, 2007; Slaughter & Greguras, 2009), challenge considered the extent to which individuals perceived a given job to offer challenging and interesting work (e.g., Carless & Imber, 2007; Turban et al., 1998), and travel referred to perceptions of travel opportunities associated with the prospective job (e.g., Van Hoye & Saks, 2011).

Within organizational characteristics, we identified five categories (organizational image, work environment, familiarity, size, location) with nine subcategories (including image, prestige/reputation, coworkers, diversity, employee relations/treatment, job security, supervisor/management, teamwork/social activities, flextime/work-life balance). Within organizational image, when predictors measured the content of perceptions that job seekers hold about an organization (Cable & Turban, 2001), such as overall company image, concern for the environment, high ethical standards, community involvement, product quality (e.g., Turban et al. 1998), organization personality perceptions (e.g., Slaughter & Greguras, 2009) and trait inferences (e.g., Lievens, Van Hoye, & Schreurs, 2005), they were included in the image subcategory. The

prestige/reputation subcategory included coefficients measuring perceptions of high status/regard for organizations (e.g., Collins, 2007). Within the work environment, the coworkers subcategory included perceptions of the anticipated type of relationship with coworkers (e.g., cooperative, close relationships; Saks, Wiesner, & Summers, 1994), and the anticipated characteristics of coworkers (e.g., warm/friendly; Carless & Imber, 2007). Diversity included perceptions regarding whether diversity was valued at the organization (e.g., Avery, Hernandez, & Hebl, 2004) as well as the status composition of gender and racial minorities within recruitment materials (e.g., Umphress, Smith-Crowe, Brief, Dietz, & Watkins, 2007). Employee relations/treatment referred to perceptions of the quality of current employee–organization relationships (e.g., Greening & Turban, 2000) and perceptions of anticipated organizational support (e.g., Casper & Buffardi, 2004). The job security subcategory included perceptions of the extent to which jobs at a prospective organization were secure (e.g., Lievens et al., 2005). Supervisor/management referred to perceptions of the quality of supervision (e.g., Slaughter & Greguras, 2009), management demands (e.g., Brown, Cober, Keeping, & Levy, 2006), and anticipated supervisor support (e.g., Turban et al., 1995). Teamwork included perceptions of whether the organization provided opportunities for teamwork (Rau & Hyland, 2003) and social/team activities (e.g., Lievens et al., 2005). Flextime/work-life balance consisted of whether the organization provided flexible hours (e.g., Kausel & Slaughter, 2011), flexible schedules (e.g., Casper & Buffardi, 2004), family-friendly practices (e.g., Bourhis & Mekkaoui, 2010), and telecommuting opportunities (e.g., Rau & Hyland, 2002). Last, familiarity reflected how familiar organizations' were to job seekers (e.g., Collins, 2007), size considered the number of employees working for an organization (e.g., Turban & Keon, 1993), and location referred to various aspects of a firm's location (e.g., Turban et al., 1998). Other characteristics that were included in the organizational characteristics category but were not placed into a subcategory included a variety of business features such as organizational policies (Bauer & Aiman-Smith, 1996), centralization (Turban & Keon, 1993), and technological development (Froese, Vo, & Garrett, 2010).

There were seven further categories generated for recruiter behaviors (competence, personableness, trustworthiness, informativeness, time recruiting, time gathering information, and structured behavior). Coefficients were classified into competence, personableness, trustworthiness, and informativeness according to the descriptions used by Barber (1998). Applicant perceptions of the percentage of time that the recruiter spent on actively recruiting and persuading them (Phillips & Dipboye, 1989) were included in the time recruiting category whereas applicant perceptions of how much time the recruiter spent gathering information (Macan &

Dipboye, 1990) were included in the time gathering information category. Coefficients in the structured behavior category measured the extent to which applicants perceived that the recruiter followed a structured format (Turban & Dougherty, 1992). Recruiter behaviors that were not placed into a subcategory included a variety of behaviors such as aggressiveness (Harris & Fink, 1987) and time spent fielding questions (Macan & Dipboye, 1990).

With respect to recruitment process characteristics, we included procedural but not distributive justice perceptions because (a) applicants experience the procedures before they know the outcomes (Truxillo, Steiner, & Gilliland, 2004); (b) procedural justice affects the entire recruiting process, whereas distributive justice only comes into play at the end of the process when job choice decisions are made; (c) most researchers have focused on procedural justice, using distributive justice only as a control or moderating variable (Ryan & Ployhart, 2000); and (d) organizations have more control over the recruitment process than the outcomes that applicants receive, so recruiters may be able to foster more favorable procedural (but not distributive) fairness perceptions (Truxillo et al., 2004). To best quantitatively summarize the primary literature on recruitment process characteristics, we used seven categories with 12 subcategories and a further three secondary subcategories. Information characteristics referred to specific qualities of the information provided during the recruitment process such as perceptions of accuracy and relevance (e.g., van Birgelen, Wetzels, & Van Doelen, 2008), adequacy (e.g., Saks et al., 1994), amount (e.g., Allen, Van Scotter, & Otundo, 2004), detail (e.g., Collins, 2007), informativeness (e.g., Thoms, Chinn, Goodrich, & Howard, 2004), and usefulness (e.g., Goldberg & Allen, 2008). Website characteristics included perceptions of website aesthetics and ease of use/navigability (e.g., Cober et al., 2003). Coefficients were included into the appeal of process subcategory when they measured applicants' affective reactions (e.g., likeability, satisfaction, attitudes) toward a variety of aspects of the recruitment process (e.g., recruitment materials, website, site visit, selection activities). The message credibility subcategory included coefficients measuring perceptions of the credibility of the recruitment source (e.g., Allen et al., 2004). Employee endorsements consisted of variables related to the utilization of employee testimonials or endorsements in recruitment-related activities (e.g., Collins, 2007; Walker, Feild, Giles, Armenakis, & Bernerth, 2009). The categorization of justice variables were based on Gilliland's (1993) model of applicant reactions. Specifically, procedural justice referred to applicant perceptions of the formal characteristics of the procedures and methods used during the process (i.e., job relatedness, predictive validity, face validity, opportunity to perform, timeliness, consistency of administration, difficulty faking, test ease) as

well as the overall fairness of the process (e.g., Bauer et al., 1998). Interactional justice included components of both informational and interpersonal justice. The former included applicants' perceptions regarding the adequacy of explanations and feedback provided to them about selection procedures and decisions, whereas the latter referred to the perceived quality of interpersonal treatment applicants received during the process (Bell, Wiechmann, & Ryan, 2006). Subcategories for interpersonal justice were consistent with Bauer, Truxillo, Paronto, Weekley, & Campion (2004) and included the quality of interpersonal treatment, propriety of questions, and two-way communication. Other recruitment process characteristics that were not placed in a subcategory included a variety of actions organizations take during the recruitment process such as sponsorship activities (Collins & Stevens, 2002) and on-campus presence (Jaidi, Van Hooft, & Arends, 2011), as well as characteristics of recruitment and selection materials/activities such as fit feedback (Dineen et al., 2002), interview focus (Chapman & Zweig, 2005), and transparency of test (Madigan & Macan, 2005).

Perceived fit was further meta-analyzed into person-job and person-organization fit categories (Kristof, 1996). Finally, perceived alternatives consisted of applicants' perceptions of the availability of alternative employment opportunities (Chapman & Webster, 2006), and hiring expectancies consisted of applicants' perceptions of their likelihood of receiving a job offer from another company (Saks & McCarthy, 2006).

Independence of Data Points

In many instances, one primary study contained data on multiple relationships that were coded into the same meta-analysis. In addition, many studies were longitudinal in nature and included measurements of the same relationship at multiple points in time (e.g., Bauer et al., 1998). To minimize the distortions that multiple coefficients from the same study can produce, we used two techniques. First, whenever a single study contributed multiple coefficients to a population estimate measured at the same point in time, we distributed the sample size among the coefficients within each phase to minimize overweighting a given sample (Steel & Kammeyer-Mueller, 2002). Second, because one of the goals of this study was to examine the effects of recruitment stage as a potential moderating variable, when primary studies contained coefficients on the same relationship from multiple stages, all coefficients were used (see Hausknecht et al., 2004, for use of this procedure). We therefore report both k_s (indicating the number of studies that contained coefficients on a relationship) as well as k_c (representing the total number of coefficients included in the

analysis) so that the number of studies and the number of coefficients are clearly elucidated.

Computation of Population Effect Sizes

We used the psychometric meta-analytic techniques outlined by Hunter and Schmidt (1990) to estimate the population effect sizes (r_c) for each relationship. Sample-size weighted average correlation coefficients were computed for each relationship with k_s equal to, or greater than, two. Each mean weighted relationship was then corrected for sampling error and for unreliability in both the predictor and criterion measures. When reliability information was not reported in a primary study, artifact distributions were used to correct for unreliability of the measures (Hunter & Schmidt, 1990; Schmidt & Hunter, 2001). In addition, we computed 95% confidence intervals to assess the accuracy of the estimated mean of the population parameters (Whitener, 1990).

Moderator Detection and Estimation

Although the vast majority of recruitment studies have focused on the first two stages, we coded for all three of Barber's (1998) stages. Both the predictor and criterion of each relationship were coded as to the stage at which they occurred. At the first stage, participants in the study were simply prospective applicants who had not yet completed a formal job application (i.e., the "generating applicants" stage). At the second stage, applicants had officially entered the applicant pool and had completed interactions with the hiring organization but had not yet received an offer (i.e., "the maintaining applicant status" stage). At the third stage, applicants had received an offer of employment on which they were making (or had already made) a decision regarding whether to accept or reject the offer (i.e., the "influencing applicant decisions" stage; Dineen & Soltis, 2011).

We used the Q statistic (Hedges & Olkin, 1985) as a test of effect size heterogeneity. When a significant Q statistic indicated heterogeneity, we conducted three subgroup analyses. First, we examined study design (laboratory vs. field) wherein laboratory studies were conducted in a laboratory generally with a student sample, whereas field studies included prospective job seekers who were actively engaged in the recruitment/job search process. Second, we examined data source of the recruitment predictor (self-report vs. other) based on whether the predictor was reported by the applicant or from a second source (such as an objective indicator). Third, we explored whether publication status of the study (published vs. unpublished) could account for heterogeneity in the data.

Results

Table 1 depicts the meta-analytic relationships between the seven recruitment predictors and applicant attraction. Note that only the 95% confidence intervals between perceived alternatives and the time recruiting and time gathering information subcategories of recruiter behaviors as well as test ease subcategory of recruitment process characteristics with applicant attraction included zero; all other relationships are significant. Table 2 displays the meta-analytic relationships between the seven recruitment predictors with job-choice behaviors. The confidence intervals for both perceived fit and perceived alternatives with job-choice behaviors included zero and are therefore deemed nonsignificant. All other recruitment predictors were significant predictors of job choice behavior. Table 3 depicts zero-order meta-analyses among the recruitment predictor categories. All relationships except recruiter behaviors with fit and hiring expectancies, and fit with perceived alternatives, are significant.

Stage as a Moderator of the Recruiting Predictor to Applicant Attraction Relationship

Table 4 depicts the meta-analytic relationships between the recruiting predictors and applicant attraction across the recruiting stages.⁴ We analyzed both concurrent measurements of the predictors and applicant attraction as well as multiple time combinations for predictive measurements. For the purpose of hypothesis testing, we focus on concurrent measurement as it provides for the clearest delineation of recruitment stage.

Meta-analytic subgroup analyses were used to explore Hypothesis 1, which predicted that the strength of the relationship between perceived alternatives and characteristics of the job, organization, and recruitment process with applicant attraction would be stronger at later as opposed to earlier recruitment stages. As predicted, recruitment process characteristics is a progressively stronger predictor of applicant attraction as the stages advance, organizational characteristics is a stronger predictor of applicant attraction at Stage 2 than Stage 1, and perceived alternatives becomes a significant predictor at Stage 2 having been a nonsignificant predictor at Stage 1. The relationship between job characteristics and applicant attraction, however, did not change significantly as a function of recruitment stage. Together, these findings provide partial support for Hypothesis 1.

⁴Stage was not assessed as a potential moderator for the job choice criterion because job choice behaviors are measured at the third recruitment stage.

TABLE 1
Zero-Order Meta-Analytic Correlations Between Recruitment Predictors and Applicant Attraction

	<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_c</i>)	<i>N</i>	<i>Q</i>
Job characteristics	0.30	0.36²¹ (0.33 – 0.39)	56 (208)	12,855	517.28**
Total compensation	0.24	0.29(0.23 – 0.34)	35 (61)	7,931	360.83**
Compensation/pay/salary	0.20	0.23(0.16 – 0.31)	29 (47)	6,134	340.60**
Benefits	0.29	0.31(0.13 – 0.48)	6 (9)	1,924	157.05**
Job itself	0.35	0.42(0.39 – 0.45)	33 (121)	7,414	166.83**
Autonomy	0.20	0.22(0.15 – 0.29)	3 (4)	1,057	4.94
Advancement/promotions	0.29	0.35(0.31 – 0.40)	14 (29)	4,022	46.77**
Challenge	0.39	0.46(0.41 – –0.51)	6 (22)	2,994	54.99**
Development	0.42	0.49(0.25 – 0.73)	2 (2)	240	7.37**
Travel	0.25	0.29(0.15 – 0.43)	4 (4)	1,045	17.72**
Organizational characteristics	0.31	0.36² (0.34 – 0.38)	116 (571)	28,514	1,621.69**
Organizational image	0.41	0.48(0.45 – 0.51)	69 (265)	17,978	851.84**
Image	0.39	0.45(0.42 – 0.48)	54 (208)	14,354	657.21**
Prestige/reputation	0.45	0.53(0.48 – 0.59)	31 (57)	7,481	318.15**
Work environment	0.25	0.30(0.26 – 0.34)	58 (164)	14,953	855.99**
Coworkers	0.45	0.31(0.23 – 0.37)	7 (21)	1,996	39.00**
Diversity	0.10	0.12(0.07 – 0.17)	19 (29)	4,028	57.41**
Employee relations/treatment	0.49	0.58(0.52 – 0.63)	15 (31)	2,953	79.14**
Job security	0.21	0.25(0.15 – 0.36)	9 (9)	2,016	40.83**

continued

TABLE 1 (continued)

	Applicant attraction			
	<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_e</i>)	<i>N</i> <i>Q</i>
Supervisor/management	0.19	0.22(0.08 – 0.37)	5 (6)	941 24.68**
Teamwork/social activities	0.32	0.37(0.25 – 0.50)	6 (6)	1,635 35.04**
Flextime/work-life balance	0.11	0.12(0.05 – 0.19)	5 (7)	1,541 13.75*
Familiarity	0.21	0.24(0.20 – 0.29)	26 (54)	8,182 186.78**
Size	0.08	0.09(0.02 – 0.15)	4 (5)	1,217 6.92*
Location	0.18	0.22(0.15 – 0.29)	15 (41)	3,784 142.33**
Recruiter behaviors	0.25	0.31³(0.28 – 0.34)	33 (211)	7,079 238.04
Competence	0.23	0.27(0.22 – 0.32)	11(43)	2,712 64.51*
Personableness	0.32	0.38(0.33 – 0.42)	26(74)	5,790 197.19**
Trustworthiness	0.22	0.30(0.12 – 0.48)	3(7)	190 6.43
Informativeness	0.15	0.19(0.12 – 0.25)	13(37)	3,146 69.18**
Time recruiting	–0.02	–0.02(–0.13 – 0.09)	2(5)	223 3.37
Time gathering information	–0.01	–0.01(–0.06 – 0.04)	2(5)	223 0.72
Structured behavior	–0.10	–0.13(–0.18 – –0.08)	2(5)	543 1.15
Recruitment process characteristics	0.24	0.29³(0.27 – 0.31)	118 (737)	47,773 2,251.78**
Information characteristics	0.10	0.12(0.08 – 0.16)	21(80)	22,873 538.73**
Website characteristics	0.36	0.41(0.37 – 0.46)	19(46)	5,238 145.18**
Aesthetics	0.34	0.39(0.30 – 0.47)	11(15)	2,512 68.04**
Navigability/ease of use	0.37	0.41(0.36 – 0.47)	17(31)	4,918 133.26**
Appeal of process	0.38	0.45(0.39 – 0.50)	32(68)	11,222 558.45**
Message credibility	0.30	0.35(0.25 – 0.44)	11(18)	3,196 129.43**
Employee endorsements	0.19	0.22(0.16 – 0.30)	3(5)	1,110 5.32

continued

TABLE 1 (continued)

	Applicant attraction				Q
	R	r_c (CI)	$k_s(k_c)$	N	
Procedural justice	0.20	0.25(0.21 – 0.27)	60(323)	28,574	1,421.29**
Overall perceptions	0.34	0.41(0.37 – 0.45)	47(118)	15,715	660.64**
Job-relatedness	0.14	0.17(0.14 – 0.21)	15(53)	3,072	38.63
Predictive validity	0.21	0.25(0.18 – 0.31)	5(8)	1,155	7.12
Face validity	0.30	0.38(0.31 – 0.45)	5(10)	2,728	20.09*
Opportunity to perform	0.24	0.30(0.27 – 0.33)	15(46)	4,907	40.49
Consistency	0.27	0.32(0.27 – 0.37)	8(34)	1,555	29.75
Free from bias	0.27	0.32(0.19 – 0.45)	4(10)	1,036	39.18**
Timeliness	0.21	0.24(0.19 – 0.30)	10(32)	1,717	33.83
Difficulty of faking	0.15	0.20(0.07 – 0.32)	4(7)	821	13.90*
Test ease	0.12	0.15(–0.15 – 0.44)	4(5)	1,286	117.34**
Interactional justice	0.32	0.40(0.37 – 0.44)	36(131)	13,374	459.17**
Informational justice	0.32	0.42(0.36 – 0.50)	18(40)	7,992	256.17**
Interpersonal justice	0.31	0.36(0.31 – 0.40)	23(75)	7,508	296.45**
Interpersonal treatment	0.37	0.43(0.37 – 0.48)	17(48)	5,637	191.64**
Propriety	0.21	0.25(0.18 – 0.34)	7(13)	1,225	22.14*
Two-way communication	0.21	0.25(0.15 – 0.37)	5(14)	1,849	62.65**

continued

TABLE 1 (continued)

	Applicant attraction			
	<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_c</i>)	<i>N</i> <i>Q</i>
Perceived fit	0.55	0.63[†](0.60 – 0.66)	43 (121)	410.50^{**}
Person–job	0.52	0.59(0.53 – 0.65)	13(35)	134.56 ^{**}
Person–organization	0.55	0.63(0.59 – 0.67)	39(86)	403.09 ^{**}
Hiring expectancies	0.21	0.25[‡](0.21 – 0.30)	20 (59)	89.16^{**}
Perceived alternatives	0.04	0.05[‡](–0.10 – 0.18)	7 (19)	104.33^{**}

Note. *r* = mean weighted coefficient. *r_c* = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. *k_s* = number of studies. *k_c* = number of coefficients. *N* = total sample size. *Q* = *Q* statistic. **P* < 0.05. ** *P* < 0.01. For the recruiting predictor categories: 1 = the superscript 1 indicates the largest coefficient, 2 = significantly smaller coefficients than those with the superscript 1 as indicated by nonoverlapping confidence intervals but larger than those with superscript 3, 3 = significantly smaller coefficients than those with the superscript 2 but larger than superscript 4. †Job characteristics is not a significantly larger predictor than recruiter behaviors. The *k_s*, *k_c*, and *N* for the subcategories (e.g., total compensation and job itself) may not total its overarching category (e.g., job characteristics). Specifically, there were instances of overlapping predictor variables (e.g., compensation and advancement–job pursuit intentions, Turban et al., 1998). In this instance, the coefficient was included in the job characteristics category and omitted from the total compensation, compensation/pay/salary, job itself, and advancement/promotions analyses, respectively. Moreover, there were instances where a predictor variable/coefficient could not be placed into a specific subcategory and/or *k_s* = 1, however, the coefficient was retained for the overarching category analysis. For example, dress code was included in the work environment and organizational characteristic analyses, but did not have a separate subcategory analysis as *k_s* = 1. Alternately, applicant perceptions of recruiters “aggressiveness” were included in recruiter behaviors and omitted from all of the recruiter behavior subcategories.

TABLE 2
Zero-Order Meta-Analytic Correlations Between Recruitment Predictors and Job Choice Decisions

		JC	OC	RB	RPC	FIT	HE	PA
Job-choice decision	<i>R</i>	0.23	0.11	0.12	0.15	0.05	0.19	0.02
	<i>r_c</i> (CI)	0.25 (0.13–0.38)	0.12 (0.07–0.17)	0.13 (0.08–0.18)	0.18 (0.16–0.20)	0.06 (–0.13–0.23)	0.20 (0.10–0.30)	0.03 (–0.14–0.20)
<i>k_s</i> (<i>k_c</i>)		4 (11)	5 (8)	12 (28)	2 (6)	5 (6)	3 (3)	
<i>N</i>		751	722	7,533	118	523	423	
<i>Q</i>		4.43	3.67	20.92	2.79	7.18	7.65	

Note. *r* = mean weighted coefficient. *r_c* = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. *k_s* = number of studies. *k_c* = number of coefficients. *N* = total sample size. *Q* = *Q* statistic. * *P* < 0.05. JC, OC, RPC, RB, FIT, HE, PA refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, perceptions of fit, hiring expectancies, and perceived alternatives, respectively. Corrected coefficients in bold are statistically significant.

TABLE 3
Zero-Order Meta-Analytic Correlations Among Recruitment Predictors

Recruitment predictors		OC	RB	RPC	FIT	HE	PA
JC	<i>r</i>	0.23	0.22	0.16	0.36	0.15	-0.06
	<i>r_c</i> (CI)	0.28 (0.26-0.31)	0.28 (0.23-0.32)	0.20 (0.14-0.26)	0.44 (0.37-0.52)	0.18 (0.11-0.27)	-0.07 (-0.10-0.05)
	<i>k_s</i> (<i>k_c</i>)	46 (513)	11 (72)	20 (69)	8 (20)	7 (22)	6 (11)
OC	<i>N</i>	13,258	2,245	6,654	2,045	1,140	3,104
	<i>Q</i>	880.89*	58.70	353.15*	46.14*	30.93	3.66
	<i>r</i>	0.23	0.23	0.41	0.36	0.18	0.12
RB	<i>r_c</i> (CI)	0.29 (0.25-0.33)	0.29 (0.25-0.33)	0.50 (0.48-0.53)	0.42 (0.37-0.47)	0.21 (0.14-0.29)	0.17 (0.11-0.22)
	<i>k_s</i> (<i>k_c</i>)	16 (121)	16 (121)	56 (311)	28 (95)	12 (25)	3 (6)
	<i>N</i>	3,826	3,826	34,313	6,296	2,194	1,629
RPC	<i>Q</i>	138.70	138.70	726.90*	354.50*	63.21*	6.99
	<i>r</i>	0.37	0.37	0.37	0.05	0.24	0.04
	<i>r_c</i> (CI)	0.47 (0.41-0.52)	0.47 (0.41-0.52)	0.47 (0.41-0.52)	0.07 (-0.17-0.29)	0.29 (0.24-0.34)	0.04 (-0.04-0.12)
RPC	<i>k_s</i> (<i>k_c</i>)	15 (92)	15 (92)	15 (92)	2 (5)	12 (35)	2 (4)
	<i>N</i>	3,050	3,050	3,050	841	2,221	703
	<i>Q</i>	141.69*	141.69*	141.69*	40.38*	34.53*	3.37
RPC	<i>r</i>	0.28	0.28	0.28	0.28	0.26	0.08
	<i>r_c</i> (CI)	0.33 (0.28-0.37)	0.33 (0.28-0.37)	0.33 (0.28-0.37)	0.33 (0.28-0.37)	0.33 (0.23-0.42)	0.11 (0.04-0.18)
	<i>k_s</i> (<i>k_c</i>)	20 (76)	20 (76)	20 (76)	20 (76)	10 (20)	6 (16)
Q	<i>N</i>	4,967	4,967	4,967	4,967	2,156	3,192
	<i>Q</i>	170.67*	170.67*	170.67*	170.67*	71.76*	32.75*

continued

TABLE 3 (continued)

Recruitment predictors	OC	RB	RPC	FIT	HE	PA
FIT	r					0.04
	$r_c(\text{CI})$					0.04 (-0.08-0.16)
	$k_s(k_c)$					2 (3)
	N					477
HE	Q					5.40
	r					0.17
	$r_c(\text{CI})$					0.20 (0.06-0.34)
	$k_s(k_c)$					4 (5)
	N					584
	Q					10.55*

Note. r = mean weighted coefficient. r_c = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. k_s = number of studies. k_c = number of coefficients. N = total sample size. Q = Q statistic. * $P < 0.05$. JC, OC, RPC, RB, FIT, HE, PA refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, perceptions of fit, hiring expectancies, and perceived alternatives, respectively.

TABLE 4
Predictive and Concurrent Phases Moderator Analyses on Applicant Attraction

Timing (predictor–outcome)	JC	OC	RB	RPC	FIT	HE	PA
Stage 1–Stage 1							
<i>r</i>	0.29	0.31	0.29	0.17	0.55	0.19	0.39
<i>r_c</i> (CI)	0.34 (0.30–0.39)	0.35 ² (0.32–0.37)	0.34 ¹ (0.27–0.41)	0.20 ³ (0.16–0.23)	0.63 (0.58–0.67)	0.23 (0.14–0.33)	0.41 (–0.03–0.85)
<i>k_s</i> (<i>k_c</i>)	40 (122)	92 (421)	4 (24)	62 (225)	34 (75)	6 (12)	2(2)
<i>N</i>	9,836	22,852	1,147	34,339	7,542	1,283	375
<i>Q</i>	497.88*	1,291.72*	29.94	2,150.18*	447.78*	27.37*	52.94*
Stage 1,2–Stage 2							
<i>r</i>	0.33	0.36	0.25	0.28	0.55	0.21	
<i>r_c</i> (CI)	0.39 (0.35–0.42)	0.43 ¹ (0.38–0.47)	0.31 ¹ (0.27–0.34)	0.34 ² (0.32–0.35)	0.65 (0.61–0.69)	0.25 (0.20–0.31)	
<i>k_s</i> (<i>k_c</i>)	18 (78)	28 (126)	28 (160)	60 (489)	10 (40)	15 (43)	
<i>N</i>	3,807	6,702	5,766	17,352	2,469	2,894	
<i>Q</i>	82.06	366.01*	209.23*	541.35*	45.48*	71.11*	
Stage 2–Stage 2							
<i>r</i>	0.33	0.38	0.28	0.29	0.56	0.21	–0.08
<i>r_c</i> (CI)	0.40 (0.36–0.44)	0.46 ¹ (0.41–0.50)	0.34 ¹ (0.30–0.37)	0.35 ² (0.33–0.37)	0.66 (0.61–0.70)	0.26 (0.20–0.31)	–0.10 ¹ (–0.13–0.06)
<i>k_s</i> (<i>k_c</i>)	17 (66)	26 (106)	27 (159)	56 (455)	10 (34)	15 (40)	5(17)
<i>N</i>	3,579	5,360	16,159	2,469	2,894	1,193	
<i>Q</i>	154.85*	334.51*	169.75*	519.32*	45.02*	71.87*	5.07

continued

TABLE 4 (continued)

Timing (Predictor-Outcome)	JC	OC	RB	RPC	FIT	HE	PA
Stage 1,2,3 – Stage 3	r	0.22	0.34	0.14	0.37		
	$r_c(CI)$	0.25 (0.04–0.46)	0.38 (0.29–0.46)	0.17 ² (0.15–0.20)	0.49 ¹ (0.46–0.53)		
	$k_s(k_c)$	2 (3)	3 (6)	3 (7)	8 (37)		
	N	226	351	256	4,214		
Stage 3– Stage 3	Q	6.75*	3.41	0.37	43.85		
	r				0.37		
	$r_c(CI)$				0.50 ¹ (0.46–0.55)		
	$k_s(k_c)$				7 (24)		
	N				4,096		
	Q				39.37*		

Note. r = mean weighted coefficient. r_c = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. k_s = number of studies. k_c = number of coefficients. N = total sample size. Q = Q statistic. JC, OC, RPC, RB, FIT, HE, PA refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, perceptions of fit, hiring expectancies, and perceived alternatives, respectively. * $P < 0.05$. Within each column, 1 = those coefficients with the superscript 1 are the largest coefficients for that predictor, 2 = significantly smaller coefficients than those with the superscript 1 as indicated by nonoverlapping confidence intervals but larger than those with superscript 3.

Hypothesis 2 predicted that recruiter behaviors would be a weaker predictor of applicant attraction at later as opposed to earlier recruitment stages. Inspection of Table 4 reveals that recruiter behaviors is not a significantly weaker predictor of applicant attraction at Stage 2 than it was at Stage 1. The predictive measurement of recruiter behaviors at Stage 3 indicates that recruiter behaviors are less predictive later than they were earlier; however, there are not concurrent data for recruiter behaviors with applicant attraction at Stage 3. Hypothesis 2 received limited support.

Our research question asked whether recruitment stage would moderate the relationship between perceptions of fit and hiring expectancies with applicant attraction. Table 4 reveals no differences in the strengths of relationships between fit or hiring expectancies with applicant attraction between Stage 1 and Stage 2. Therefore, it would appear that fit perceptions are the strongest predictor, and hiring expectancies are significant but one of the smallest recruitment predictors of applicant attraction during both the generating applicants and maintaining applicant status stages.

Relative Strength of Recruitment Predictors

Meta-analytic confidence intervals were used to explore Hypothesis 3, which predicted that job characteristics, organizational characteristics, and perceptions of fit would be stronger predictors of applicant attraction and job choice behaviors than the recruitment process and recruiter behaviors. Using Cohen's (1988) recommendations to interpret the population effect sizes (i.e., that an r of at least 0.10 be termed a small effect, an r of at least 0.24 be termed a medium effect, and an r of at least 0.37 be termed a large effect), inspection of Table 1 reveals that perceived fit is the largest predictor (and a large effect) of applicant attraction averaged across recruitment stage. As evidenced by nonoverlapping confidence intervals, organizational characteristics and job characteristics are the next largest predictors (significantly smaller than perceived fit and medium-sized effects); organizational characteristics was a significantly stronger predictor than all of the remaining four predictors, and job characteristics was significantly larger than all but recruiter behaviors. The third largest predictors of applicant attraction were recruiter behaviors, recruitment process characteristics, and hiring expectancies, which were not different from each other and all medium-sized effects. Finally, perceived alternatives was a nonsignificant predictor of applicant attraction. With the exception that job characteristics was not a significantly stronger predictor of applicant attraction than recruiter behaviors, Hypothesis 3 was supported for the applicant attraction outcome.

TABLE 5
Incremental Variance Accounted for on Applicant Attraction by Each Recruiting Predictor

Variable	<i>B</i>	<i>sr</i> ²	ΔR^2 restricted	Order	ΔR^2 step
Model $R^2_{AA(JCOCRBPCFIT)} = 0.500$, $F(5,4042) = 809.55$, $P < 0.01$					
$r^2_{AA(JC.OCRCRBFIT)}$	-0.018	0.000	0.130**	5	0.000
$r^2_{AA(OC.JCRCRBFIT)}$	0.042	0.001**	0.130**	4	0.001**
$r^2_{AA(RB.JCOCRPCFIT)}$	0.355	0.084**	0.096**	2	0.096**
$r^2_{AA(RPC.JCOCRBFIT)}$	-0.111	0.007**	0.084**	3	0.006**
$r^2_{AA(FIT.JCOCRPCRB)}$	0.610	0.245**	0.397**	1	0.397**

Note. *B* = regression coefficient in full model. sr^2 = semipartial correlation squared in full model. ΔR^2 Restricted = the variance accounted for by the predictor category beyond null model. Order = order in which predictors are entered from null model to full model. ΔR^2 Step = the variance accounted for by the predictor category when it is added to the model in stepwise regression. JC, OC, RPC, RB, FIT refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, and perceptions of fit, respectively. The notation for each effect follows that used by Pedhazur (1997). For instance, $r^2_{AA(FIT.JCOCRPCRB)}$ represents the incremental effect of perceptions of fit above and beyond job characteristics, organizational characteristics, recruitment process characteristics, and recruiter behaviors on applicant attraction.

** $P < 0.01$.

With respect to the job choice criterion, inspection of Table 2 reveals that the confidence intervals overlap between all significant predictors. Hypothesis 3 was not supported for the job-choice behavior outcome.

Incremental Variance of Recruitment Predictors

To examine the unique variance accounted for by applicant attraction on the predictor categories, we used semipartial correlations created through multiple regression on the meta-analytically derived coefficients (see Table 5). The absence of an intercorrelation estimate between hiring expectancies with perceived fit meant that hiring expectancies could not be included in the regression model. Perceived alternatives was also not included in this analysis because of its nonsignificant zero-order relationship with applicant attraction. Examination of the semipartial correlations in Table 5 reveals that all of the recruitment predictors except job characteristics account for significant amounts of unique variance in applicant attraction. Furthermore, recruiter behaviors account for the largest and recruitment process characteristics account for the second largest amounts of unique variance in applicant attraction above and beyond perceptions of fit. These results support Hypothesis 4.

Table 6 displays the regression coefficients and squared semipartial correlations between the recruitment predictors and applicant attraction

TABLE 6
Incremental Variance Accounted for on Applicant Attraction by Job, Organizational, and Recruitment Process Characteristics, Recruiter Behaviors, and Perceptions Fit by Stage of the Recruitment Process

Variable	<i>B</i>	<i>Sr</i> ²	ΔR^2 restricted	Order	ΔR^2 step
<i>Measured at Stage 1</i>					
Model $R^2_{AA(JCOCRB RPCFIT)} = 0.561, F(5,3322) = 850.29, P < 0.01$					
$r^2_{AA(JC.OCRPCRB FIT)}$	−0.076	0.004**	0.116**	4	0.004**
$r^2_{AA(OC.JCRPCRB FIT)}$	0.069	0.003**	0.122**	5	0.003**
$r^2_{AA(RB.JCOCRPC FIT)}$	0.474	0.149**	0.116**	2	0.116**
$r^2_{AA(RPC.JCOCRB FIT)}$	−0.282	0.047**	0.040**	3	0.042**
$r^2_{AA(FIT.JCOCRPCRB)}$	0.727	0.331**	0.397**	1	0.397**
<i>Measured at Stage 2</i>					
Model $R^2_{AA(JCOCRB RPCFIT)} = 0.566, F(5,3385) = 881.26, P < 0.01$					
$r^2_{AA(JC.OCRPCRB FIT)}$	0.008	0.000	0.160**	5	0.000
$r^2_{AA(OC.JCRPCRB FIT)}$	0.140	0.013**	0.212**	3	0.010**
$r^2_{AA(RB.JCOCRPC FIT)}$	0.338	0.076**	0.116**	2	0.116**
$r^2_{AA(RPC.JCOCRB FIT)}$	−0.087	0.004**	0.122**	4	0.005**
$r^2_{AA(FIT.JCOCRPCRB)}$	0.626	0.245**	0.436**	1	0.436**

Note. *B* = regression coefficient in full model. *sr*² = semipartial correlation squared in full model. ΔR^2 Restricted = the variance accounted for by predictor category beyond null model. Order = order in which predictors are entered from null model to full model. ΔR^2 Step = the variance accounted for by the predictor category when it is added to the model in stepwise regression. JC, OC, RPC, RB, FIT refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, and perceptions of fit, respectively. The notation for each effect follows that used by Pedhazur (1997). For instance, $r^2_{AA(FIT.JCOCRB RPC)}$ represents the incremental effect of perceptions of fit above and beyond job characteristics, organizational characteristics, recruiter behaviors, and recruitment process characteristics on applicant attraction.

***P* < 0.01.

at the generating applicants and maintaining applicant status stages of the recruitment process. Of note, there were too few coefficients to conduct an analysis of incremental variance for Stage 3. Inspection of the order of entry of predictor categories into the model at Stage 1 versus Stage 2 in Table 5 indicates that stage of the recruitment process is a significant moderator of the relationships between the recruitment predictors and applicant attraction. Perceived fit is the strongest predictor and recruiter behaviors account for the next largest amount of incremental variance in applicant attraction at both Stages 1 and 2. Further, and consistent with Hypothesis 1, organizational characteristics is more predictive at the second stage of the recruitment process. However, the unique influence of recruitment process characteristics is lower and for job characteristics is nonsignificant at the second stage of the recruitment process. Consistent with Hypothesis 4, inspection of Table 6 reveals that recruiter behaviors is

TABLE 7
*Incremental Variance Accounted for on Job-Choice by Each
 Recruiting Predictor*

Variable	<i>B</i>	<i>Sr</i> ²	ΔR^2 Restricted	Order	ΔR^2 Step
Model $R^2_{\text{JOBC(JCOCRB(RPCHE))}} = 0.096, F(5,1605) = 33.940, P < 0.01$					
$r^2_{\text{JOBC(JC.OCRPCRBHE)}}$	0.213	0.199**	0.063**	1	0.063**
$r^2_{\text{JOBC(OC.JCRPCRBHE)}}$	-0.018	-0.015	0.014**	4	0.000
$r^2_{\text{JOBC(RB.JCOCRPCHE)}}$	-0.014	-0.012	0.017**	4	0.000
$r^2_{\text{JOBC(RPC.JCOCRBHE)}}$	0.109	0.085**	0.032**	3	0.008**
$r^2_{\text{JOBC(HE.JCOCRPCRB)}}$	0.234	0.129**	0.040**	2	0.025**

Note. *B* = regression coefficient in full model. *sr*² = semipartial correlation squared in full model. ΔR^2 Restricted = the variance accounted for by the predictor category beyond null model. Order = order in which predictors are entered from null model to full model. ΔR^2 Step = the variance accounted for by the predictor category when it is added to the model in stepwise regression. JC, OC, RPC, RB, HE refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, and hiring expectancies, respectively. The notation for each effect follows that used by Pedhazur (1997). For instance, $r^2_{\text{AA(HE.JCOCRPCRB)}}$ represents the incremental effect of hiring expectancies above and beyond job characteristics, organizational characteristics, recruitment process characteristics, and recruiter behaviors on applicant attraction.

** $P < 0.01$.

less uniquely influential on applicant attraction at the second stage (7.6%) than the first stage of the recruitment process (14.9%).

For job-choice decisions, we used semipartial correlations created through multiple regressions on the meta-analytically derived coefficients to examine the unique variance accounted for by job-choice behavior on the predictor categories (see Table 7). Perceived fit and alternatives were not used in this analysis because of their nonsignificant zero-order relationships with job choice behaviors. Examination of the semipartial correlations revealed that only job characteristics, recruiter behaviors, and hiring expectancies accounted for significant amounts of unique variance in job choice. Job characteristics contributed the most unique variance followed by hiring expectancies and recruiter behaviors. These results partially support Hypothesis 4.

Moderator Testing

Tables 8 through 10 display subgroup meta-analyses for the recruitment predictors with applicant attraction by laboratory and field study designs, source of the recruitment predictor data (self-report vs. other), and publication status, respectively. Inspection of Table 8 reveals that although job characteristics were larger predictors of applicant attraction in field designs as compared to laboratory designs, recruitment process characteristics, recruiter behaviors, hiring expectancies, and perceived

TABLE 8
*Zero-Order Meta-Analytic Correlations Between the Predictors and Applicant
 Attraction Moderated by Study Design*

		Applicant attraction				
		<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_c</i>)	<i>N</i>	<i>Q</i>
JC	Lab	0.26	0.32 ² (0.27 – 0.37)	35 (99)	7,713	389.82*
	Field	0.36	0.42 ¹ (0.39 – 0.45)	21 (109)	5,142	92.81
OC	Lab	0.30	0.35(0.33 – 0.38)	89 (367)	22,117	1,469.47*
	Field	0.32	0.39(0.36 – 0.41)	27 (204)	6,397	181.6
RPC	Lab	0.37	0.42 ¹ (0.40 – 0.45)	71 (325)	19,278	912.88*
	Field	0.16	0.20 ² (0.18 – 0.22)	47 (412)	28,496	877.88*
RB	Lab	0.33	0.40 ¹ (0.33 – 0.45)	9 (37)	1,698	46.53
	Field	0.23	0.29 ² (0.25 – 0.32)	24 (174)	5,538	185.76
FIT	Lab	0.56	0.64(0.60 – 0.67)	33 (91)	7,523	403.49*
	Field	0.51	0.62(0.57 – 0.66)	10 (30)	2,230	37.21
HE	Lab	0.28	0.34 ¹ (0.27 – 0.42)	9 (21)	1,501	44.86*
	Field	0.17	0.21 ² (0.16 – 0.25)	11 (38)	2,561	34.80
PA	Lab	0.30	0.32 ¹ (0.06 – 0.57)	3 (8)	440	57.66*
	Field	-0.07	-0.09 ² (-0.13 – -0.09)	4 (11)	1,128	3.85

Note. *r* = mean weighted coefficient. *r_c* = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. *k_s* = number of studies. *k_c* = number of coefficients. *N* = total sample size. *Q* = *Q* statistic. **P* < 0.05. JC, OC, RPC, RB, FIT, HE, PA refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, perceptions of fit, hiring expectancies, and perceived alternatives, respectively. For each recruitment category, those coefficients with the superscript 1 are significantly larger than coefficients with the superscript 2 as indicated by nonoverlapping confidence intervals.

alternatives were all stronger effects in the lab than in the field. Study design did not produce an effect size difference for perceived fit or for organizational characteristics with applicant attraction.

With respect to the recruitment source moderator, all of the measures of applicant attraction were self-report in nature. Table 9 reveals that when the recruitment predictor source was also self-report, the correlations with applicant attraction were higher across all predictors than when the recruitment predictor rating came from another source.

Table 10 reveals two significant findings for publication status. Specifically, recruitment process characteristics produced larger effects in published than unpublished studies whereas the opposite was found for recruiter behaviors (i.e., the relationship between recruiter behaviors and attraction was larger for unpublished than published studies).

Discussion

The purposes of this paper were to explore whether the relationships between recruitment predictors and outcomes change as a function of

TABLE 9
Zero-Order Meta-Analytic Correlations Between the Predictors and Applicant Attraction Moderated by Source of the Recruitment Predictor

		Applicant attraction				
		<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_c</i>)	<i>N</i>	<i>Q</i>
JC	Self-report	0.35	0.42 ¹ (0.39 – 0.44)	46 (182)	10,785	238.69*
	Other	0.09	0.09 ² (0.01 – 0.18)	11 (26)	2,721	149.81*
OC	Self-report	0.36	0.42 ¹ (0.40 – 0.44)	97 (502)	23,674	1, 327.86*
	Other	0.13	0.14 ² (0.11 – 0.17)	26 (69)	7,639	133.91*
RPC	Self-report	0.34	0.41 ¹ (0.39 – 0.42)	104 (667)	32,864	1, 147.20*
	Other	0.05	0.06 ² (0.03 – 0.09)	24 (70)	16,766	255.33*
RB	Self-report	0.26	0.32 ¹ (0.29 – 0.35)	31 (201)	6,899	242.39*
	Other	0.18	0.20 ² (0.15 – 0.24)	4 (10)	529	2.08

Note. *r* = mean weighted coefficient. *r_c* = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. *k_s* = number of studies. *k_c* = number of coefficients. *N* = total sample size. *Q* = *Q* statistic. **P* < 0.05. JC, OC, RPC, RB refer to job characteristics, organizational characteristics, recruitment process characteristics, and recruiter behaviors, respectively. For each recruitment category, those coefficients with the superscript 1 are significantly larger than coefficients with the superscript 2 as indicated by nonoverlapping confidence intervals. Several primary studies included both self-report and other-report data for the predictors and thus, the *k_s* will sum to greater than the total *k_s* for analyses containing these primary studies.

recruitment stage, and to identify the recruiting predictor(s) that are most meaningful for applicants in predicting their attraction to organizations, and to identify whether predictors account for significant incremental variance in applicant attraction and job choice behavior when considered in combination.

With respect to applicant attraction, perceived fit clearly emerges as the strongest predictor of applicant attraction as a zero-order predictor and incrementally to other predictors across the recruitment stages. Although job and organizational characteristics are the second largest zero-order predictors of applicant attraction (*r_c* = 0.36), neither predictor accounts for much of the unique variance. Organizational characteristics was a stronger predictor at Stage 2 than at Stage 1 in terms of zero-order correlations, and it accounted for more incremental variance at Stage 2 (1.3%) than Stage 1 (0.3%). There was no difference in the zero-order predictive strength of job characteristics across the stages, and it accounted for only 0.4% of the incremental variance in applicant attraction at Stage 1 and none at Stage 2.

The zero-order relationship between recruiter behaviors and applicant attraction was tied for third largest (*r_c* = 0.31) overall. Nevertheless, recruiter behaviors accounted for the greatest proportion of the incremental variance after fit at Stage 1 (14.9%), and at Stage 2 (7.6%). Recruitment process characteristics was a stronger zero-order predictor of applicant

TABLE 10
*Zero-Order Meta-Analytic Correlations Between the Predictors and Applicant
 Attraction Moderated by Publication Status*

		Applicant attraction				
		<i>R</i>	<i>r_c</i> (CI)	<i>k_s</i> (<i>k_c</i>)	<i>N</i>	<i>Q</i>
JC	Published	0.28	0.34(0.30 – 0.37)	39 (155)	9,378	373.20*
	Unpublished	0.35	0.41(0.36 – 0.47)	17 (53)	3,477	132.18*
OC	Published	0.28	0.34(0.31 – 0.36)	72 (357)	19,337	1,002.01*
	Unpublished	0.36	0.41(0.38 – 0.45)	44 (214)	9,177	579.16*
RPC	Published	0.32	0.38 ¹ (0.36 – 0.40)	80 (501)	27,597	1,001.05*
	Unpublished	0.14	0.17 ² (0.13 – 0.20)	38 (236)	20,177	946.05*
RB	Published	0.22	0.27 ² (0.24 – 0.30)	28 (201)	5,764	179.42
	Unpublished	0.39	0.49 ¹ (0.36 – 0.60)	5 (10)	1,472	42.02*
FIT	Published	0.49	0.57(0.49 – 0.65)	14 (36)	2,822	162.28*
	Unpublished	0.58	0.65(0.62 – 0.68)	29 (85)	6,931	202.99*
HE	Published	0.19	0.23(0.19 – 0.28)	16 (50)	3,137	63.77
	Unpublished	0.27	0.32(0.22 – 0.43)	4 (9)	925	21.01*
PA	Published	0.05	0.05(–0.09 – 0.19)	6 (15)	1,416	102.85*
	Unpublished†					

Note. *r* = mean weighted coefficient. *r_c* = coefficient corrected for sampling error and unreliability of the predictor and criterion. CI = 95% confidence interval. *k_s* = number of studies. *k_c* = number of coefficients. *N* = total sample size. *Q* = *Q* statistic. **P* < 0.05. JC, OC, RPC, RB, FIT, HE, PA refer to job characteristics, organizational characteristics, recruitment process characteristics, recruiter behaviors, perceptions of fit, hiring expectancies, and perceived alternatives, respectively. For each recruitment category, those coefficients with the superscript 1 are significantly larger than coefficients with the superscript 2 as indicated by nonoverlapping confidence intervals. †There were not sufficient primary studies to conduct a meta-analysis between PA and applicant attraction for unpublished studies.

attraction as the stages advanced, albeit it accounted for more incremental variance at Stage 1 (4.7%) than Stage 2 (0.4%). Hiring expectancies was a medium-sized effect in terms of the zero-order relationship with applicant attraction (*r_c* = 0.25), which did not change as a function of recruitment stage. Perceived alternatives was a nonsignificant predictor of applicant attraction at Stage 1 but was a significant negative predictor of applicant attraction at Stage 2.

With respect to job choice, as evidenced by overlapping confidence intervals in Table 2, none of the predictors emerged as the strongest zero-order predictor. Regression analysis results displayed in Table 7 further revealed that, whereas job characteristics accounted for the most unique variance followed by hiring expectancies and then recruitment process characteristics, organizational characteristics and recruitment process characteristics do not account for unique variance in job-choice decisions.

This meta-analysis differs from the contributions made in Chapman et al. (2005) in five distinct ways. First, the Chapman et al. (2005)

meta-analysis focused on modeling the relationships between recruiting outcomes (job/organization attraction, intention, and job choice behaviors). In this manuscript, our focus is on the interrelationships between the predictors including relative strength and incremental variance of the predictors on applicant attraction. As such, our search for samples proceeded with a different set of search terms and also required a recruitment process characteristics category that included information about the appeal and usability of recruitment activities (Cober et al., 2003), reactions to recruiting messages (Breau et al., 2008), and reactions to the recruitment process (Ryan & Ployhart, 2000). Through the incremental variance analyses, we can begin to understand some of the conflicting earlier findings relating recruiter behaviors to applicant attraction (e.g., Carless & Imber, 2007; Rynes & Miller, 1983); although recruiter behaviors does not have a particularly large zero-order relationship with applicant attraction, it accounts for a significant proportion of the incremental variance above and beyond the other predictors.

Second, we examined how relationships between the recruitment predictors and applicant attraction change as a function of stage of the recruitment process. Through examining the data by stage, we see that organizational characteristics and recruitment process characteristics are weighed more heavily, whereas recruiter behaviors are weighed less as the stages progress.

Third, we conducted subgroup analyses by study design (laboratory vs. field), source of the recruitment information (self-report vs. other), and publication status (published vs. unpublished). These moderator results are discussed below.

Fourth, whereas Chapman et al. (2005) included 70 studies collected to the middle of 2002, this study included 232 studies and 9 further years of recruitment research. The greater number of primary studies allowed for more specificity in the categories, and as such, we included 36 novel meta-analyses to those used by Chapman et al. In examining effect sizes for comparable analyses, notable differences include that our estimate of the variance accounted for in applicant attraction by perceived fit is larger, and variance accounted for by hiring expectancies is lower. Also, in their meta-analysis, Chapman et al. (2005) found an unanticipated direction in the relationship between perceived alternatives and job/organization attraction wherein as perceived alternatives went up, attraction went up as well. By looking at perceived alternatives across the stages, we see that perceived alternatives is a significant and negative predictor at the maintaining applicants stage ($r_c = -0.10$; such that as perceived alternatives go down, attraction goes up), but it is not significant earlier at the application decision stage.

Fifth, when comparing the results from Chapman et al. to this study with respect to job choice behaviors, several observations are noteworthy. First, whereas Chapman et al. (2005) did not find organizational characteristics to be a significant predictor of job choice ($r_{xy} = 0.07$), it was a significant predictor in this study ($r_c = 0.12$), which may be due to the greater number of coefficients now available (11 vs. 6). Second, although Chapman et al. did not find that their perceptions of the recruitment process category significantly predicted job choice ($\rho = 0.09$), the recruitment process characteristics category was a significant predictor in this study ($r_c = 0.18$). The recruitment process characteristics category used here included some additional aspects to the Chapman et al. category and a substantially larger number of coefficients (28 vs. 3). Third, the coefficients relating job characteristics and recruiter behaviors to job choice are much larger in this study ($r_{cs} = 0.25$ and 0.13 for job characteristics and recruiter behaviors in this study vs. 0.10 and 0.10 in Chapman et al.). Fourth, there were no additional coefficients that examined the relationship between perceived fit and job choice between Chapman et al. and this study. Although this relationship is not statistically significant, it is based on only six coefficients from two studies and a total sample size of 118. Before strong conclusions can be drawn about the relationship between perceived fit and job choice, we suggest that additional research on larger samples is warranted.

Theoretical Implications

Although Kristof (1996) predicted that perceived fit might play a larger role in organizations' decisions once overall qualifications are established toward the end of the selection process, our findings suggest that perceived fit plays a large role in applicant attraction throughout the recruitment process. This suggests that applicants generate assessments of fit throughout the recruitment process. Of note, the zero-order correlations in Table 1 suggest that both person-organization and person-job fit are important considerations in fit assessments. That said, what the applicants value fit to—the job or the organization—may change as a function of recruitment stage. Albeit from the recruiter's perspective in a selection context, Chaung and Sackett (2005) found that that recruiters may attend more to person-job fit at the initial interview stage and person-organization fit at the final interview stage. Chuang and Sackett (2005) contend that person-job fit may not become less important, but rather the variability in person-job fit has been reduced for the final set of applicants, and person-organization fit is the remaining differentiator. Carless (2005) found that both person-job and person-organization fit predicted attraction, but only person-job fit accounted for unique variance in acceptance intentions.

Disentangling the effects of fit with different aspects of a position (e.g., the job and the organization) to determine if these forms may differentially predict various outcomes (Bretz & Judge, 1998; Bretz, Rynes, & Gerhart, 1993; Cable & DeRue, 2002) is an area that is ripe for future research.

Consistent with the ELM, image theory, and signaling theory, characteristics of the recruitment process are stronger zero-order predictors of applicant attraction as the recruitment process progresses. Although this may suggest that organizations extend greater resources toward the recruitment process later in recruiting, its unique added variance is only 0.3% in the second stage of recruitment versus 4.7% at Stage 1. This is not to suggest that recruitment process characteristics become less important relative to other predictors at later stages. Rather, the variability in recruitment process characteristics is reduced in later stages of the recruitment process, thereby leaving other predictors as the primary differentiators of applicant attraction at later stages. A similar observation was made by Chuang and Sackett (2005) with respect to person-job fit in a selection context.

We found some evidence to suggest that recruiter behaviors matter more at earlier recruitment stages in that they account for a substantial proportion of the incremental variance at early stages of the recruitment process as compared to later stages. Examining the zero-order relationships did not reveal the importance of recruiter behaviors in accounting for unique variance in applicant attraction as compared to job and organizational characteristics. One potential explanation is that perhaps information about job and organizational characteristics is used to inform perceptions of fit. Recruiter behaviors, on the other hand, may be considered by applicants separately from fit perceptions. Conceptual work examining whether perceived fit might mediate the relationship between job and organizational characteristics with applicant attraction seems warranted (Uggerslev et al., 2009). Similarly, these findings highlight the importance of examining the recruiting predictors in tandem. Because recruitment practices are rarely employed in isolation, it is important to understand the impact of each recruitment practice within a recruitment system.

Our follow-up moderator analysis of study design type suggests that laboratory designs may be artificially inflating the importance of recruiter behaviors, recruitment process characteristics, hiring expectancies, and perceived alternatives as compared to naturally occurring recruitment contexts. Perhaps we are designing our stimuli and cues too well or to be overly salient in our laboratory studies as compared to field settings. And lab designs may be underestimating the importance of job characteristics. At the end of a real job search, the hire must actually show up, do the work, and

be paid what has been set out. It does not seem surprising that asking mock applicants in the laboratory about jobs at which they do not have to actually work would produce less salient effects than asking actual applicants in field settings (see also Breaugh, 2008). Because job characteristics did not account for incremental variance above and beyond the other predictors and because study design was a significant moderator, field designs are warranted to draw conclusions about the role of job characteristics in predicting attraction. In addition, more research is needed to understand the role of job characteristics in job-offer-acceptance decisions.

The moderator analysis regarding the source of the recruitment predictor rating indicated that predictor relationships are potentially inflated when applicants are asked to rate both attractiveness of the job characteristics, organizational characteristics, recruitment process characteristics, or recruiter behaviors, and to provide ratings of their attraction. Although this suggests single-source bias, within the recruitment context it is not what *recruiters identify* as attractive that matters but rather what *applicants perceive* as attractive that matters. Consequently, researchers may be justified in their use of applicants to rate attractiveness of recruitment predictors. Thus, to minimize single source bias, perhaps the focus should be on alternate sources (i.e., not self-report) for the criterion. Objective measures of actual job-choice decisions would be optimal, and research is sorely needed examining job acceptance behaviors. Although increased attention is being paid to job-choice decisions, for instance, a whole chapter by Harold et al. in *Oxford's Handbook of Recruiting* (Cable & Yu, in press), we echo the calls that many previous researchers have made for more job-choice behavior research (e.g., Barber, 1998; Breaugh et al., 2008; Dineen & Soltis, 2011). Researchers' focus on the first two stages of the recruitment process (Allen et al., 2007; Van Hove & Lievens, 2009) using predominantly attitudinal and intentions-based outcomes may perhaps be at the expense of behavioral data from the influencing job-choice decisions stage. Nonetheless, examining applicant behavior does not have to wait until the third stage where job acceptance decisions are made. Applicant behaviors at the earlier two stages involve application decisions and observations of applicants' decisions to continue in the applicant pool or to withdraw (Harold et al., in press).

Practical Implications

Fit emerging as the largest predictor of applicant attraction across recruitment stages generates the question of what organizations can do to foster perceptions of fit. Kristof-Brown, Zimmerman, and Johnson (2005) contended that there is little research that has been conducted on the antecedents of perceived fit. Dineen et al. (2002) explored providing

applicants with feedback on the extent of their fit to an organization and found that fit feedback will foster higher subjective fit and attraction. Another area that holds promise toward fostering fit is to tailor the recruitment process to each applicant according to the applicant's information preferences (e.g., Kraichy, Uggerslev, & Fassina, 2009). Within the finite time that recruiters spend with applicants during the recruiting process, focusing on aspects of the job or organization that are key for each applicant might improve applicant attraction (Newman & Lyon, 2009) and perceptions of fit. For instance, one candidate may not be interested in childcare and parental leave benefits but keenly interested in bus routes and social aspects of the workplace. Future research is needed to explore whether tailoring information content to applicant preferences and configural customization of when recruitment information is provided may foster higher fit perceptions (Dineen & Noe, 2009). It is worth noting that, of course, recruiters need to be ethical in emphasizing areas of true fit without portraying the organization as something it is not (Buckley, Fedor, Carraher, Frink, & Marvin, 1997).

In addition to potentially cultivating higher fit perceptions, organizations may wish to target applicants who already have a compatible list of values and goals. Kristof (1996) suggested that because they illustrate an organization's specific goals and values, realistic job previews and site visits might promote higher levels of fit than more general strategies such as newspaper advertisements (Rynes, 1991). Specific efforts must be undertaken to make organizational values salient to applicants, so recruitment sources that provide greater amounts of information (e.g., recruitment brochures) may allow candidates with different sets of goals and values to self-select out of the recruitment process.

In terms of Behling et al.'s (1968) implicit theories, the strongest predictors of applicant attraction collapsed across the multiple stages of the recruitment process are subjective in nature (i.e., fit). Although predictors that are objective in nature (relating to job and organizational attributes) have the second largest zero-order relationships to applicant attraction, critical contacts (such as recruiters and the recruitment process) account for the most incremental variance next to fit. Practically speaking, the results of this study would suggest that organizations should direct their initial recruitment resources at fostering applicants' perceptions of fit. Incremental recruitment resources should be invested in recruiters—knowing that their behaviors will have an impact in particular for early applicant attraction, with investments to follow in organizational characteristics—knowing that these account for incremental variance in applicant attraction during the maintaining applicant status stage, and the recruitment process—knowing that it becomes a stronger predictor of attraction as the stages progress.

This pattern of results brings some good news for practitioners as organizations may only be able to feasibly improve some of the categories of recruiting information (Yuce & Highhouse, 1998). Whereas it may be challenging to change organization size, location, the work itself, and organizational image, it may be feasible to train recruiters on how to behave during interviews, develop attractive web application tools, and enhance recruitment procedures. The categories of predictors that organizations have a great deal of control over (recruiter behaviors and characteristics of the recruitment process) may account for different variance than what is accounted for by fit perceptions. Although some previous research has questioned the contribution of recruiters to applicant attraction, the incremental variance results presented here suggest that investment in these areas would not be wasted. Most notably, personableness was a significantly larger predictor than any of the other recruiter behaviors. Practitioners can now select or train for recruiter personableness knowing that it makes a significant and unique contribution to applicant attraction.

The results of this study have implications for the longstanding debate about the difference between statistical and practical significance. The stepwise regression order and associated ΔR^2 indicate that all recruitment activities except job characteristics predict applicant attraction, although job characteristics account for the most unique variance in job choices. Recruitment specialists, however, must consider the potential return on investment for each recruitment dollar spent and decide whether it is worth expending limited resources beyond fostering fit and training recruiters if other predictors are only going to result in a small improvement.

A closer inspection of the subcategory results from Table 1 reveals several practical implications from this study. First, with respect to job characteristics, total compensation seems to matter less than the job itself; applicants appear to care about more than simply being "shown the money." Within organizational characteristics, image is a stronger zero-order predictor than specific characteristics of the organization. Together, these findings suggest that what motivates a candidate to enter an applicant pool and eventually accept a position may not be the same as what keeps the employee at the organization. For example, Griffeth, Hom, and Gaertner (2000) note that the strongest predictors of turnover are job satisfaction, leadership, and stress. Second, personableness was the recruiter behavior that most strongly predicted applicant attraction. This suggests that perhaps being friendly is the most important recruiter behavior.

Study Limitations

As we have defined them, different characteristics of the recruitment process may come into play at earlier versus later stages. For instance,

the appeal and usability of early recruitment activities such as job ads and websites (Walker, Feild, Giles, Armenakis, & Bernath, 2009) may be useful in setting up heuristics early in the process. At later stages in the process, applicants will have more information and be more motivated to make an effortful evaluation of fair treatment during the recruitment process (Barber, 1998).

Of note, as reflected in Table 4 a paucity of research has explored the relationships between recruitment predictors and applicant attraction at the influencing job-choice stage of the recruitment process. This is consistent with the explicit shift to focusing on the earlier stages of the recruitment process given that, if applicants do not apply for positions, they disappear from later recruitment and selection activities (Allen et al., 2007; Carlson, Connerley, & Mecham, 2002; Van Hove & Lievens, 2009). Although the first two stages of the recruitment process may be where recruitment information is most informative and when organizations have the greatest flexibility in manipulating recruitment information (Barber, 1998), there may be a role for recruiters, recruitment activities, or for job and organizational characteristics to influence applicants' job-choice decisions. For instance, behaviors of a recruiter may be particularly salient to applicants at later stages if the recruiter is a potential supervisor or coworker. In addition, recruiting activities and attempts to foster fit at the job-choice decision stage may be particularly important with respect to posthire outcomes such as tenure (Newman & Lyon, 2009), job satisfaction, organizational commitment, and performance (Zottoli & Wanous, 2000) given their temporal proximity.

Inspection of Tables 5 and 6 reveals the presence of negative beta weights associated with job and recruitment process characteristics suggesting that both serve a net negative suppression function (Cohen & Cohen, 1975; Conger 1974; Horst, 1941; Krus & Wilkenson, 1986). More specifically, job and recruitment process characteristics suppress the error variance thereby increasing the predictive validity of the remaining three predictors with applicant attraction (e.g., Darmawan & Keeves, 2006). The suppression of error variance in combination with the strong intercorrelations between job characteristics and recruitment process characteristics with the remaining three predictor variables generates the negative beta weights. This statistical artifact, however, limits practitioners' ability to interpret the potential positive incremental contribution of job and recruitment process characteristics with applicant attraction.

A further limitation of these meta-analyses stems from collapsing narrow predictor categories (e.g., favorability of the work itself, compensation, and advancement) into broad predictor categories (e.g., job characteristics) for our phase and incremental variance analyses. Although our interrater agreement for the collapse of narrow predictors into broader

predictor categories was extremely high and this procedure has been used by others (e.g., Chapman et al., 2005), this collapsing does lead to a loss in specificity of the relationships and may mask nuanced differences in relationships between narrow categories. This collapse was necessary to obtain a sufficient number of coefficients so that broad predictor conclusions could be drawn (such as the relationship between the job characteristics category and indices of applicant attraction) and so that moderators (such as recruitment stage) could be examined. A closer inspection of the subcategories in Table 1 reveals many similarities in effect sizes between the subcategories and the broader categories. Likewise, although we collapsed job pursuit, acceptance intentions, and job/organizational attraction into one applicant attraction outcome, and therefore lost some specificity, this collapse substantially increased the number of primary studies included in each meta-analysis, thereby improving the relationship estimate.

Additional Future Research

Although the ELM is a useful theory to explain the mechanisms through which recruiter behaviors and recruitment process characteristics versus job and organizational characteristics may predict applicant attraction (i.e., through invoking heuristics versus through effortful scrutiny of the information; Petty & Briñol, 2008) and that effortful scrutiny will be more persuasive than heuristics in predicting attraction, in this study we have not tested the mechanism used by applicants at each stage. Furthermore, we did not test whether heuristics may produce similar sized effects as compared to effortful scrutiny in successive steps of the recruitment process. That is, there may be a similar magnitude of relationship between a heuristic and applicant attraction at an earlier stage as compared to between a careful scrutiny of evidentiary merit and applicant attraction at a later stage. To thoroughly understand how the mechanisms of persuasion may change over the course of the recruitment process and to understand the persistence of attitudes formed through effortful scrutiny of information versus through heuristics, longitudinal research is sorely needed.

Numerous authors have examined various conceptualizations and operationalizations of perceived fit (e.g., Caplan, 1987; Edwards, 1991; Kristof, 1996; Muchinsky & Monahan, 1987). Most of the primary studies used in this meta-analysis define perceived fit essentially as supplementary fit, wherein a person “supplements, embellishes, or possesses characteristics which are similar to other individuals” (Muchinsky & Monahan, 1987, p. 271). Measurement of complementary fit—the extent to which an individual possesses characteristics that make them a missing puzzle piece—are much less common (Kristof, 1996). If measures of fit

perceptions are developed to measure both fit conceptualizations, we may be able to enhance our prediction of applicant attraction with perceived fit.

From the organization's perspective, understanding the scope of consequences of their recruitment and selection activities is critically important and has not been researched thoroughly enough. For instance, what is the effect of information learned about an organization by an individual who is not currently engaged in the job search process? Will it affect the person's future job search behaviors? Similarly, what is the impact of news media, product marketing and branding efforts, and personal experience with or exposure to an organization and its products? Related to these questions, Allen et al. (2007) found that organization image is related to job pursuit intentions after controlling for information obtained during a subsequent job search. Willness, Uggerslev, and Chapman (2009) conceptually explored how image about an organization derived from product use and previous exposure to the organization may generalize or spillover onto future recruitment drives. Collins (2007) found that low-information recruitment practices, such as recruitment posters and sponsorship at public venues, were significantly related to application behaviors through employer familiarity and employer reputation when applicants had low familiarity with the organization's products. In contrast, high-information recruitment practices focusing on job and organizational attributes were related to job seekers' application behaviors through employer reputation and job information when the organization's products were well-known to the applicant. There is also much room for further understanding of optimal timing for recruitment activities. For instance, when should site visits be conducted, early on to give applicants a realistic job preview, or at later stages when meeting the prospective workgroup might be more informative for applicants' job acceptance decisions? When should applicants optimally be given information about the compensation package and work environment? As the war for talent reaches new heights in the next decade, these variables and their timing may lead to an edge over the competition or an uphill battle depending on their perceived favorability.

Conclusion

This paper examined the relative strength and incremental variance of seven recruiting predictors on applicant attraction and job choice, and explored the potential for differences in relationship magnitudes across different stages of the recruiting process. The theoretical implications stemming from the results of this study include that applicants may value different types of information about an opportunity and be persuaded

through different mechanisms depending on the stage of the recruitment process. The practical implications of this study suggest that recruiters should spend their first recruitment dollar focusing on fostering perceptions of fit through message content. Incremental recruitment dollars should be spent on enhancing recruiter behaviors at the early stages of recruiting, fostering favorable recruitment processes, and on promoting favorable organizational characteristics to foster attraction at later recruitment stages.

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